

LEONARDO FABYAN ORTEGA RIVERA

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EDUCATION

Universidad de Guadalajara

Guadalajara, Jalisco, Mexico

Bachelor of Science in Computer Engineering (GPA: 91.42/100)

November 2022 – December 2026 (Expected)

- Completed 100% of coursework; modular graduation in progress. Academic focus on Machine Learning, Tensor Mathematics, Systems Programming, and Numerical Methods.

Institución Universitaria de Envigado

Envigado, Colombia

Scientific Research Exchange — Programa DELFÍN

June 2025 – July 2025

- Architected relational database schema (OUL-DB) to catalog multi-spectral Flexible Image Transport System (FITS) astronomical image arrays and satellite telemetry; engineered indexing pipelines for computational astrophysics modeling.

EXPERIENCE

Power Industrial

Guadalajara, Jalisco, Mexico

Software & Systems Developer

August 2024 – Present

- Engineered desktop inventory management system in Python and SQLite3, reliably managing 3,600+ industrial SKUs across local warehouse operations.
- Architected multi-workstation local network (LAN) concurrency across Windows network shares via SQLite in WAL mode with busy timeout handling, eliminating database lock contention for active client stations.
- Enforced strict schema validation and transactional integrity to eliminate data corruption; automated scheduled database backups and lightweight client auto-update workflows.

SELECTED RESEARCH & ENGINEERING PROJECTS

Inside Deep Learning | *PyTorch, torch.einsum, Analytical Jacobians, Mathematical Derivations*

Active Development

- Derived and implemented backpropagation, multi-dimensional tensor representations, and exact analytical Jacobian derivatives from mathematical first principles without PyTorch Autograd.
- Leveraged Einstein summation notation (`torch.einsum`) for tensor contractions and vector-Jacobian products (VJPs); achieved 0.00% Mean Absolute Percentage Error (MAPE) validation against native PyTorch functional layers.

DeepLense Gravitational Lensing Classifier | *PyTorch, ResNet-18, PINN, Grad-CAM, Optuna*

2026

- Fine-tuned modified ResNet-18 computer vision backbone for dark matter subhalo detection in strong lensing images; executed Bayesian optimization (Optuna), achieving 93.76% accuracy and 0.9908 One-vs-Rest ROC-AUC.
- Architected Physics-Informed Neural Network (PINN) variant via Radial Symmetry Residuals (RSR), boosting ROC-AUC to 0.9915; validated mass anomaly focus via Grad-CAM and proved CNN necessity via manifold clustering.

Temporal Heterogeneity in Cognitive Architectures | *Reproduction — Elsevier CSR*

2026

- Formally reproduced Sandoval-Arrayga et al. (Elsevier, 2024), measuring the evolutionary fitness impact of multiscale temporal processing ($a = 3$, up to +151.8% advantage) in embodied cognitive agents across 4,500 generations.
- Audited mathematical formulation, uncovering critical arithmetic errors in publication formulas (0.0833% vs 0.833%) documented in formal errata (`typos.md`); designed and executed 12-run parallel ablation suite (~10.6h compute).

Recurrent Convolutional Neural Networks | *PyTorch, Conv2dGRU, RCNN2d, 5D Tensors*

2026

- Engineered and benchmarked spatiotemporal models (RCNN2d, Conv2dGRU, and CNN baseline) for multi-frame video prediction on Moving MNIST using 5D tensors (`seq_len`, `batch`, `C`, `H`, `W`).
- Formulated multi-objective loss combining MSE and Structural Similarity Index (SSIM) for perceptual sharpness; implemented autoregressive decoding with decaying teacher forcing schedules.

RecycleNet MLOps Pipeline | *MobileNetV3, Astral uv, GitHub Actions CI/CD, Mypy, Bandit*

2026

- Engineered automated vision pipeline using MobileNetV3 transfer learning across 6 waste categories, integrating deterministic CUDA seeding and multi-class One-vs-Rest ROC-AUC evaluation suite.
- Implemented production CI/CD lifecycle using Astral uv, strict Mypy static typing, Ruff formatting/linting, and Bandit SAST vulnerability security scanners via automated GitHub Actions.

Dynamic NeRFs for Exospheric Reconstruction | *Latin Lunar Lobby Congress — U. Complutense de Madrid*

2026

- Formulated dynamic continuous temporal parameterizations to model non-rigid diurnal exospheric particle density shifts for the OUL satellite mission, overcoming static 3D NeRF reconstruction limitations.
- Integrated satellite orbital line-of-sight (LOS) ray-tracing simulations for optically thin Lyman- α atmospheric emissions using computational geometry and PyVista.

TECHNICAL SKILLS

Languages:

Python 3, C++ (C++17 for Algorithms & Data Structures), SQL, Bash

ML & Mathematics:

PyTorch, Tensor Algebra (`torch.einsum`), NumPy, Computer Vision, Scikit-Learn, MLflow, Optuna

DevOps & Systems:

Linux, Astral uv, Git, GitHub Actions (CI/CD), Strict Mypy, Ruff, Bandit (SAST), Systemd

Databases & Spoken:

SQLite3 (WAL Mode), MySQL · English (B1 Certified — UdeG), Spanish (Native)